

The Sources of Price Impact Kernel Shape: A Controlled Simulation

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Abstract

We investigate which market mechanisms are responsible for the shape of the price impact propagator kernel, using two controlled simulation environments: a Transient Impact Model (TIM) and a zero-intelligence Santa Fe limit order book (LOB). Both models are driven by a shared fundamental value process, ensuring that differences in their recovered kernels can be attributed to the price formation mechanism rather than to differences in information flow. The TIM recovers a clean, convex, monotonically decaying propagator consistent with the no-dynamic-arbitrage conditions of Gatheral (2010). The Santa Fe LOB produces a qualitatively different kernel: a brief initial non-monotonicity at short lags followed by an approximately linear decay to zero, violating both the non-increasing and convexity conditions. We attribute both violations to the absence of market intelligence in the zero-intelligence framework. The non-monotonicity arises from book momentum, a pattern that strategic agents would trade against. The failure to produce a convex decay reflects the absence of long-memory order flow autocorrelation induced by institutional metaorder splitting — the structural condition under which convex kernel decay is required for market efficiency.

1 Introduction

The price impact of a trade — the effect of order flow on the asset price at which it executes — is a central object in market microstructure theory. Empirical studies of high-frequency data consistently find that impact is not instantaneous: a trade moves the price immediately, but the price partially reverts over subsequent steps as the order book replenishes, eventually stabilizing at a level that reflects any informational content the trade carried. The dominant industry framework for capturing this behavior decomposes price into a permanent informational component and a transient mechanical component, whose decay is governed by a propagator kernel of a functional form encoding the structure of market impact.

The central question of this project is: which market mechanisms are responsible for the shape of this kernel? We use two controlled simulation environments — a Transient Impact Model (TIM) and a zero-intelligence Santa Fe limit order book (LOB) — to isolate the distinct contributions to kernel shape. Both models track an identical asset with a randomly-walking fundamental value, ensuring that their price dynamics can be compared on equal footing. The TIM imposes a specific kernel shape directly by construction. The LOB generates price endogenously from order book mechanics alone, with no strategic agents and no persistent order flow. Comparing the kernels recovered from each model allows us to dissect which features arise from order book mechanics, and which must then be explained by other factors such as investor behavior.

The natural next step is application to real high-frequency trading data. Having established in simulation which mechanisms compose the kernel shape, we aim to apply the same recovery procedure to real market data and decompose the empirically observed impact structure into its constituent sources.

2 Transient Impact Model

The Transient Impact Model (TIM) is the dominant framework for modeling price impact in quantitative finance (Bouchaud et al., 2004). In its canonical form, the asset price is written as the cumulative sum of past signed order flows, each weighted by a propagator kernel $G(\tau)$ that determines how the impact of a trade decays as a function of the lag τ since it was executed. This section describes the motivation for the kernel’s functional form, the theoretical constraints it must satisfy, and the full specification of our implementation.

Kernel choice: power-law decay

The choice of functional form for $G(\tau)$ is not arbitrary. A well-established empirical finding is that institutional investors — constrained by the market impact of their own trades — routinely split large parent orders into many small child orders, executed incrementally over hours or days (Bouchaud et al., 2009). This meta-order splitting induces a long-memory autocorrelation structure in the signed order flow sequence $\{\epsilon_t\}$: the autocorrelation function decays as $C(\ell) \sim \ell^{-\gamma}$ for some $0 < \gamma < 1$ (Lillo and Farmer, 2004). For markets to be efficient — specifically, for this persistent order flow not to create predictable price drift that arbitrageurs could exploit — the propagator must itself decay as a power law (Bouchaud et

al., 2009). An alternative such as an exponentially decaying kernel would under-correct for long-run impact and leave a systematic predictable component in returns.

The power-law kernel

$$G(\tau) = A(\tau + 1)^{-\beta} \quad (1)$$

with amplitude $A > 0$ and decay exponent $\beta > 0$ is therefore the theoretically motivated and empirically supported choice.

No-dynamic-arbitrage conditions

A valid propagator must additionally satisfy regularity conditions that preclude dynamic arbitrage strategies. Gatheral (2010) establishes that a price impact kernel is consistent with no-dynamic-arbitrage (NDA) if and only if it is (i) non-negative, (ii) non-increasing, and (iii) convex. Intuitively, a kernel that increases at any lag would imply that the impact of a past trade continues to accumulate indefinitely, providing a systematic profit opportunity to a trader who could detect the pattern. The power-law kernel $G(\tau) = A(\tau + 1)^{-\beta}$ satisfies all three conditions for any $A, \beta > 0$. These conditions serve as a benchmark against which the kernel recovered from the Santa Fe LOB is evaluated in Section 4.

Model specification

The TIM models the asset price at time t as the cumulative sum of decaying past impacts:

$$p_t = p_0 + \sum_{t'=1}^t q_{t'} \epsilon_{t'} [G(t - t') + \lambda_{t'}] + \eta_t \quad (2)$$

where G and λ define the mechanical and informational price impacts, respectively. The components are:

- p_t is the asset price at time t
- $q_{t'}$ is the impact magnitude of the trade at time t'
- $\epsilon_{t'}$ is the signed trade direction (± 1)
- $G(\tau = t - t')$ is the mechanical impact decay kernel
- $\lambda_{t'}$ is the informational impact function
- η_t is additive noise

Since the full convolution scales as $\mathcal{O}(N^2)$, the sum is truncated to a finite lookback window L in practice.

The key design choices in our implementation are as follows.

- $G(\tau) = A(\tau + 1)^{-\beta}$: mechanical impact decays as a power law, consistent with the motivation above.

- $\lambda_t = \begin{cases} C \sim \text{LogNormal}(\mu_\lambda, \sigma_\lambda) & \text{with probability } p_{\text{info}} \\ 0 & \text{otherwise} \end{cases}$

Trades carry informational content with probability p_{info} , causing a permanent, log-normally distributed price shift.

- $q_{t'} = \sigma_\nu D$, $D \sim \text{LogNormal}(\mu_V, \sigma_V)$, $\sigma_\nu^2 = \omega + \alpha_{\text{vol}} \eta_{t-1}^2 \sigma_{t-1}^2 + \beta_{\text{vol}} \sigma_{t-1}^2$
Impact magnitude scales with trade size, drawn from a lognormal distribution and modulated by a GARCH(1,1) stochastic volatility process σ_ν to introduce volatility clustering.
- $\epsilon_t = \text{sign}(x_t)$, $x_t = \rho x_{t-1} + \eta_t$, $\eta_t \sim \mathcal{N}(0, \sigma_\eta^2)$
Trade signs take values in $\{+1, -1\}$. An AR(1) autocorrelation process is applied to the informed component of order flow to capture the directional persistence associated with institutional metaorder execution.

3 Santa Fe Limit Order Book Model

Real markets do not manually inject price movements upon making a trade; they do so by consuming orders in a book. To complement the TIM, we design a second model in which price emerges endogenously from the interaction between agents and a stylized limit order book.

Rather than sharing order flow with the TIM, this model generates its own order flow endogenously. The two models are coupled instead through a shared fundamental value process: the same sequence $\{I_t, C_t, \sigma_t\}$ is supplied to both, ensuring they respond to the same informational events while differing entirely in their price formation mechanism. Any differences in their recovered kernel structures can therefore be attributed to the price formation mechanism itself, rather than to differences in the underlying information flow.

Shared Fundamental Process

The shared object is the sequence

$$\{I_t, C_t, \sigma_t\}_{t=1}^T \quad (3)$$

where $I_t \in \{0, 1\}$ is an informational flag (Bernoulli with probability p_{info}), $C_t \sim \text{LogNormal}(\mu_C, \sigma_C)$ is the shock size, and σ_t is the GARCH stochastic volatility. The informed trading direction is generated once and shared between both models:

$$\epsilon_t^{\text{inf}} = \text{sign}(x_t), \quad x_t = \rho x_{t-1} + \eta_t \cdot \mathbf{1}[I_t = 1] \quad (4)$$

The canonical fundamental shock sequence is then:

$$\Delta F_t = I_t \cdot C_t \cdot \epsilon_t^{\text{inf}} \cdot \sigma_t \quad (5)$$

Both models accumulate the same fundamental price path:

$$F_t = F_0 + \sum_{s=1}^t \Delta F_s \quad (6)$$

Manual Model Under Shared Fundamental

The TIM synthesizes its own uninformed order flow from a Poisson arrival process at rate μ_u per side, matching the uninformed arrival rate of the LOB. At each step, the net uninformed signed flow is:

$$\epsilon_t^{\text{noise}} = \text{sign}\left(N_t^{\text{buy}} - N_t^{\text{sell}}\right), \quad N_t^{\text{buy}}, N_t^{\text{sell}} \sim \text{Poisson}(\mu_u) \quad (7)$$

with ties mapping to zero to avoid a systematic directional bias. The composite sign uses the shared informed direction on informed steps:

$$\epsilon_t = \begin{cases} \epsilon_t^{\text{inf}} & \text{if } I_t = 1 \\ \epsilon_t^{\text{noise}} & \text{otherwise} \end{cases} \quad (8)$$

Price is constructed as before, with the permanent component built from the canonical shared shock sequence:

$$p_t = p_0 + p_t^{\text{mech}} + p_t^{\text{info}} + \eta_t, \quad p_t^{\text{info}} = \sum_{s \leq t} \Delta F_s \quad (9)$$

Santa Fe LOB Agent Simulation

The LOB model runs a full endogenous agent simulation at each time step:

1. **Fundamental update:** $F_t \leftarrow F_{t-1} + \Delta F_t$
2. **Limit order placement:** $N_{\text{lim}} \sim \text{Poisson}(\lambda \cdot d)$ orders arrive (where d is the number of depth levels), each placed on a random side at a level drawn from a geometric distribution, concentrating liquidity near the quote. The stationary depth profile that emerges is consistent with the Santa Fe model (Farmer et al., 2005):

$$D(n) = \frac{\lambda}{\delta} [1 - e^{-\delta n / \mu_u}] \quad (10)$$

where $n \geq 1$ is the level index from the best quote.

3. **Cancellations:** each resting unit cancels independently with probability δ per step.
4. **Informed market order:** if $I_t = 1$, one market order of volatility-scaled lognormal size is submitted in direction ϵ_t^{inf} .
5. **Gap-trading:** if $|F_t - p_t| > \theta$, a gap-sensitive informed agent submits a market order of size $\kappa \cdot |F_t - p_t|$ in direction $\text{sign}(F_t - p_t)$, where $\kappa > 0$ is the gap intensity and $\theta \geq 0$ is a minimum threshold below which no trade is triggered. In the spirit of Kyle (1985), the agent's demand scales with the level of mispricing rather than arising from a fixed trading probability, anchoring the LOB price to the fundamental through endogenous order flow rather than an exogenous price adjustment.
6. **Uninformed market orders:** $N^{\text{buy}} \sim \text{Poisson}(\mu_u)$ buy orders and $N^{\text{sell}} \sim \text{Poisson}(\mu_u)$ sell orders arrive, executed in random order against the book.

7. **Mid-price update:** the mid-price shifts by half a tick per depleted level. No exogenous recentering is applied; the fundamental anchor is provided entirely through the gap-trading mechanism in step 5.

Model Diagnostics

As a validation diagnostic, we track the deviation of each model’s observed price from the shared fundamental:

$$d_t = p_t - F_t \quad (11)$$

A well-calibrated model keeps $\mathbb{E}[|d_t|]$ small and d_t symmetric around zero. The TIM’s deviation reduces to $p_t^{\text{mech}} + \eta_t$, which is mean-zero when uninformed flow is symmetric. The LOB’s deviation depends on the interplay between order imbalance, book depth, and the gap-trading intensity κ . Figure 1 shows the simulated price paths for both models alongside the shared fundamental over one trading year. Figure 2 shows d_t for each model; the comparable magnitudes and zero-centered distributions confirm that neither model exhibits systematic drift, and that the controlled basis for kernel comparison is maintained throughout the simulation.

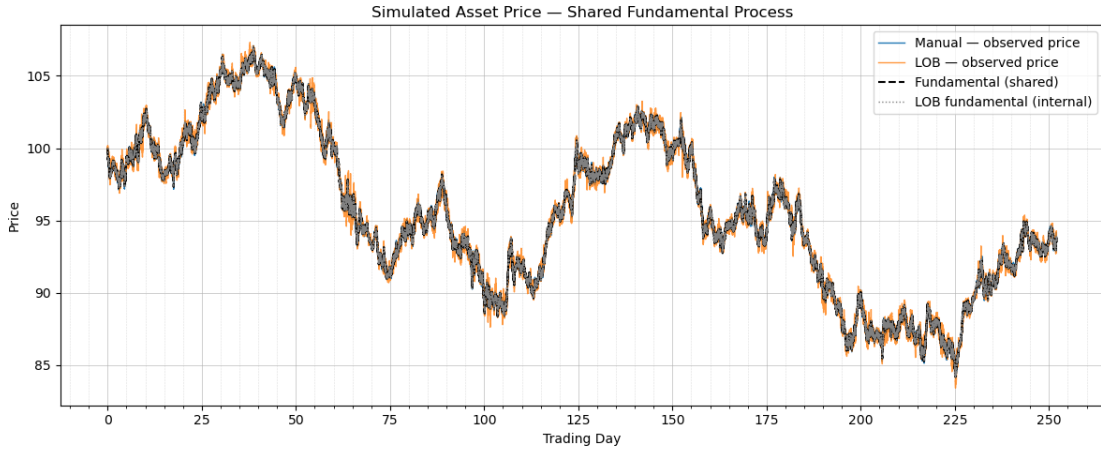


Figure 1: Simulated asset price paths for the TIM and Santa Fe LOB over one trading year, with the shared fundamental value overlaid. Both models track the same underlying process, confirming the controlled basis for kernel comparison.

4 Parameter Recovery

Recovering the propagator kernel from simulated price data is easier to accomplish in terms of returns than in terms of price levels. Consider $r_t = p_t - p_{t-1}$. Differentiating the transient impact model gives:

$$r_t = \lambda \epsilon_t q_t + \sum_{\tau \geq 1} \epsilon_{t-\tau} q_{t-\tau} [G(\tau) - G(\tau - 1)] + \eta_t \quad (12)$$



Figure 2: Deviation of each model’s observed price from the shared fundamental, $p_t - F_t$, for the TIM (top) and Santa Fe LOB (bottom). Both models maintain comparable deviation magnitudes, confirming that the controlled comparison is not compromised by systematic price drift in either model.

Equivalently, writing $K(\tau) = G(\tau) - G(\tau - 1)$ for the lag increments:

$$r_t = C\epsilon_t q_t + \sum_{\tau=1}^L K(\tau) \epsilon_{t-\tau} q_{t-\tau} + \eta_t \quad (13)$$

Returns are linear in past signed order flow, so the lag coefficients $\hat{K}(\tau)$ can be estimated by OLS regression. The propagator is then reconstructed as the cumulative sum:

$$\hat{G}(\tau) = \sum_{u=1}^{\tau} \hat{K}(u) \quad (14)$$

which is then fit to a power-law decay:

$$\hat{G}(\tau) \approx A(\tau + 1)^{-\beta} \quad (15)$$

This fitting process recovers: A (mechanical impact amplitude), β (decay exponent), and $C \cdot p_{\text{info}}$ (average informational impact per trade).

Recovery from LOB Output

When applying the regression to the LOB model’s output, the model was not generated by the TIM functional form, so the recovery does not estimate true underlying parameters.

Rather, it tests whether the power-law functional form is a useful approximation of the LOB’s effective impact structure, and — more importantly for our purposes — whether the shape of $\hat{G}(\tau)$ recovered from the LOB resembles that recovered from the TIM. The LOB does not produce a single trade sign per step; instead, we construct a net signed flow from its recorded order arrivals:

$$\epsilon_t^{\text{net}} = I_t \cdot \epsilon_t^{\text{inf}} + N_t^{\text{buy}} - N_t^{\text{sell}} \quad (16)$$

collapsed to sign $(+1, 0, -1)$ before regression, keeping the regressor on the same scale as the manual model’s ϵ_t .

Price Response Function

As a model-free complement to the regression-based recovery, we compute the *price response function*:

$$R(\ell) = \langle (p_{t+\ell} - p_t) \epsilon_t \rangle \quad (17)$$

This measures the average price change ℓ steps after a trade, signed by the direction of that trade, and requires no modeling assumptions beyond the price and signed flow series. It provides an intuitive and direct picture of how prices respond to order flow on average, without the intermediate regression step.

The relationship between $R(\ell)$ and the propagator G follows from the TIM structure. For serially uncorrelated signed flow, one can show that $R(\ell) = \langle \epsilon^2 \rangle \cdot G(\ell)$, so the response function tracks the shape of G but is scaled by the mean squared signed flow rather than normalized by it as in OLS. In practice this means $R(\ell)$ lies slightly but systematically below $\hat{G}(\tau)$: the scale factor $\langle \epsilon^2 \rangle < 1$ whenever some steps carry zero net signed flow, which occurs in both models due to symmetric Poisson arrivals occasionally cancelling. The shape, however, is preserved.

The price response function therefore serves a dual role. First, it acts as a sanity check on the regression: if $R(\ell)$ and $\hat{G}(\tau)$ disagree in shape rather than merely in level, this signals a model misspecification rather than a normalization difference. Second, when applied to the LOB, it provides a non-parametric view of the kernel shape that does not depend on the power-law functional form assumed in the regression, making it a cleaner diagnostic for comparing the two models.

5 Observations and Conclusions

The simulation performs as intended across both models. Both capture the three main stylized facts of financial returns: heavy tails at the minute and daily timescale, volatility clustering via the GARCH process, and minimal autocorrelation of returns. Figures 3 and 4 show the return distributions at the minute and daily timescales respectively; the heavier tails relative to the Gaussian overlay confirm excess kurtosis at both scales, consistent with empirically observed return distributions.

The TIM recovery performs well: A , β , and $C \cdot p_{\text{info}}$ are all recovered with reasonable accuracy. The slight systematic underestimation of A and overestimation of β is attributable to the finite lookback window L , which forces the regression to attribute impact to a shorter

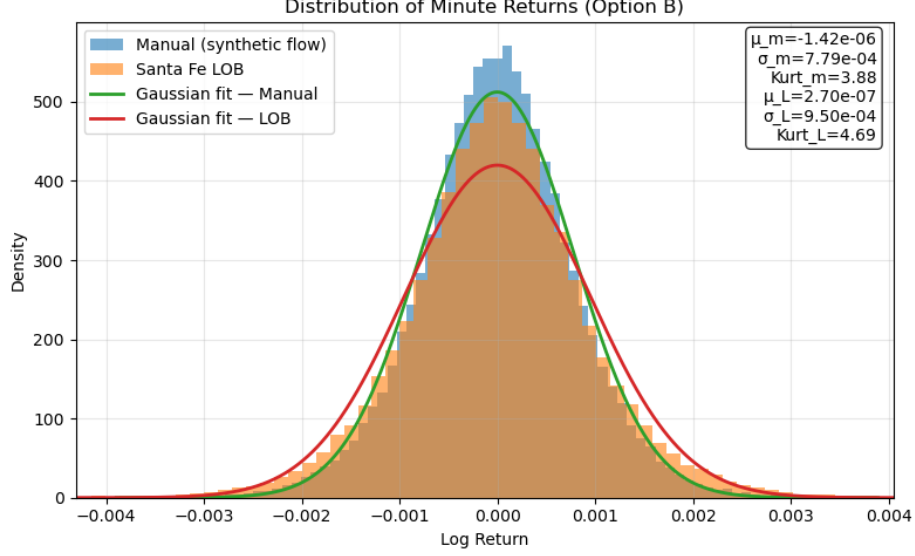


Figure 3: Distribution of minute-scale log returns for both models. Heavy tails relative to the fitted Gaussian confirm the presence of excess kurtosis at the intraday timescale.

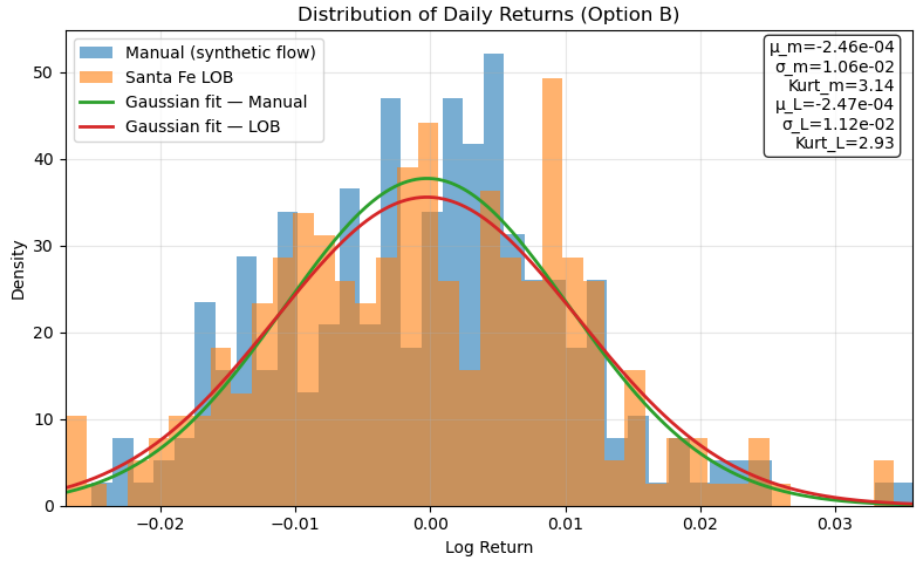


Figure 4: Distribution of daily log returns for both models, with Gaussian overlay. Volatility clustering produced by the shared GARCH process is visible in the heavier tails relative to the normal distribution.

window than the true kernel spans, producing estimates that are larger in amplitude and faster-decaying. The informational constant $C \cdot p_{\text{info}}$ is consistently overestimated by approximately 4–5%.

Kernel Shape Comparison

The central finding of this project concerns the difference in recovered kernel shape between the two models. The TIM produces a clean, monotonically decaying $\hat{G}(\tau)$ that is well-described by the imposed power-law form, with the power-law fit closely tracking both the regression estimate and the true kernel. The Santa Fe LOB produces a qualitatively different shape: $\hat{G}(\tau)$ rises sharply at the contemporaneous lag, continues to increase slightly over the first few lags before turning over, and then decays approximately linearly to zero. This result is confirmed by the non-parametric price response function $R(\ell)$, which mirrors the shape of $\hat{G}(\tau)$ in both models and establishes that the difference is not an artifact of the regression procedure. The slight but consistent gap between $R(\ell)$ and $\hat{G}(\tau)$ in both models is accounted for by the $\langle \epsilon^2 \rangle$ normalization difference discussed in Section 4 and does not affect the shape comparison. The two kernels are shown side by side in Figure 5.

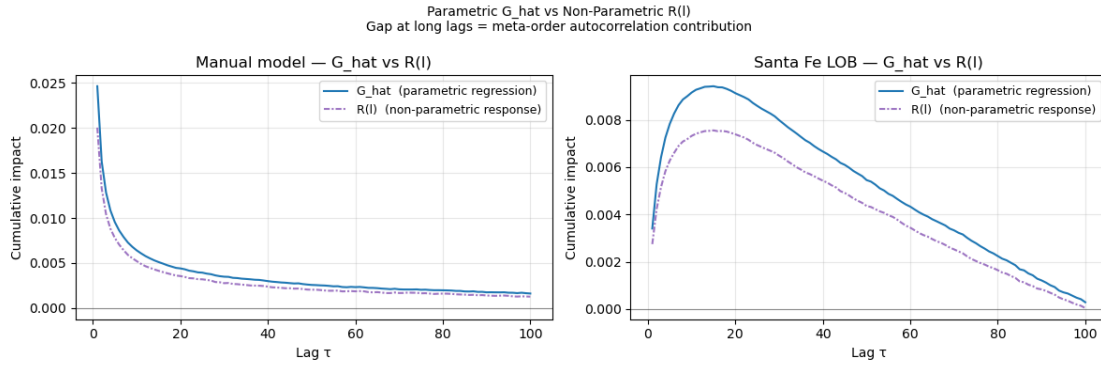


Figure 5: Recovered propagator kernel $\hat{G}(\tau)$ for the TIM (left) and the Santa Fe LOB (right), with power-law fit and non-parametric price response function $R(\ell)$ overlaid. The TIM kernel decays as a convex power law consistent with the no-dynamic-arbitrage conditions of Gatheral (2010). The LOB kernel exhibits a brief initial rise before decaying approximately linearly to zero, violating both the non-increasing and convexity conditions.

The LOB kernel violates two of the three no-dynamic-arbitrage conditions of Gatheral (2010). Non-negativity alone is satisfied: the kernel remains non-negative throughout and decays cleanly to zero. The non-increasing condition is violated at small lags, where $\hat{G}(\tau)$ rises before turning over — a brief but measurable increase implying that the impact of a past trade continues to compound over the first few steps. The convexity condition is violated at longer lags: the subsequent decay is approximately linear rather than the concave, decelerating profile of the TIM’s power law. A strictly convex kernel is one whose rate of decay slows as the lag increases; a linear kernel implies that each additional period carries the same marginal incremental impact regardless of how long ago the originating trade occurred. The power-law functional form assumed in the fit is strictly convex for all $\beta > 0$, and therefore provides a poor approximation to the LOB kernel, returning a high estimated $\hat{\beta}$ and a low R^2 .

The Role of Order Flow Autocorrelation

The non-increasing violation at small lags arises from book momentum. When a market order depletes the best levels of the ask side, the book is temporarily thinner than its stationary profile. If a subsequent order arrives in the same direction before the book has refilled, it encounters less resistance and walks proportionally further, generating a larger price shift per unit volume than the initiating order. This transient amplification of consecutive same-direction impacts produces the brief initial rise in $\hat{K}(\tau)$ before the depletion dissipates and limit order refilling restores the stationary depth profile. In a market with strategic agents, this pattern would be recognized and traded against: participants anticipating the thinned book would submit opposing orders, eliminating the predictable amplification. The zero-intelligence LOB contains no such mechanism.

The approximately linear decay of the LOB kernel is a direct consequence of the absence of long-memory order flow autocorrelation. As discussed in Section 2, the power-law shape of the TIM kernel is theoretically motivated by the requirement that, in an efficient market, the propagator must precisely cancel the directional persistence in signed flow induced by metaorder splitting. This cancellation imposes a specific convexity structure on $G(\tau)$: the kernel must decay quickly at short lags — where the autocorrelation is strongest — and decelerate at long lags as the autocorrelation fades. The result is the characteristic concave curvature of the power law.

The Santa Fe LOB contains no such autocorrelation structure. Its uninformed agents submit independent Poisson arrivals with no directional persistence, and its informed agents affect only a fraction p_{info} of steps. The aggregate signed flow is therefore approximately serially uncorrelated. With no persistent directional signal to cancel, there is no force bending the kernel toward a convex profile: the book depletion and refill mechanism produces a short-run impact that dissipates at a roughly constant rate, yielding the near-linear $\hat{G}(\tau)$ observed in the LOB output.

This is the central finding. Both NDA violations — the brief initial non-monotonicity from book momentum and the failure to produce a convex decay — are absences of market intelligence. The non-increasing violation would be eliminated by strategic agents who recognize and trade against the predictable amplification of depleted-book impacts. The convexity violation would be addressed by the long-memory autocorrelation structure that institutional metaorder splitting introduces into order flow: the kernel must curve concavely precisely because it must cancel a persistent directional signal that fades at a decelerating rate. A zero-intelligence LOB, containing neither strategic reactivity nor institutional order flow, satisfies only non-negativity. Introducing metaorder dynamics into the LOB is therefore the natural and theoretically motivated next step.

Next Steps

The immediate next step is to apply the same recovery procedure to real high-frequency data from the LOBSTER database. This is a necessary precondition for any further simulation work: if the kernel recovered from real data does not exhibit the convex, power-law decay that the TIM produces, then enriching the LOB model with additional mechanisms would have no empirical target to aim for. Confirming that the empirically observed kernel is

indeed convex and well-approximated by a power law establishes the phenomenon that the simulation must explain.

Given that confirmation, the natural next step is to introduce metaorder splitting into the LOB. Rather than a single informed market order per informed step, informed agents would execute a persistent directional campaign spanning many consecutive steps — a stylized representation of an institutional investor working a large parent order. This injects long-memory autocorrelation into the order flow, which theory identifies as the condition under which a convex, power-law kernel is required for market efficiency. However, it is important to note that the autocorrelated flow alone does not mechanically produce the convex shape. The power-law kernel is the form the market must take *in response to* persistent institutional flow in order to remain efficient — it is the pricing behavior of strategic liquidity providers, who adjust quotes and withdraw depth upon detecting autocorrelated activity, that shapes the propagator. A zero-intelligence LOB, whose limit order providers refill the book at a fixed mechanical rate regardless of the flow they face, lacks this response. Adding metaorders to the LOB therefore tests a more targeted question: how much of the kernel shape is attributable to the mechanical price reversion that follows the end of a metaorder campaign, and how much requires genuine strategic repricing? The gap between the LOB kernel with metaorders and the empirically observed power law would isolate the contribution of market intelligence specifically.

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Code availability. All simulation code is available at <https://github.com/atpower89/Market-Impact-Simulation>.